

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

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1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #67. The comparison assesses whether bug fixes or model changes affected results.

Current Comparison:

- **Issue:** #67
- **Baseline:** Commit f3674e8 (2025-11-20 09:42:40)
- **Fixed:** Commit d5d1356 (2025-11-20 11:20:03)

1.1 Available Snapshot Pairs

Table 1: All Available Snapshot Pairs

	Issue	Title	Baseline Commit	Fixed Commit	Status
3	#67	—	f3674e8	d5d1356	Current
1	#62	—	d046c7c	f3674e8	
2	#64	—	8681d1e	8681d1e	

2 Metadata Comparison

Table 2: Metadata Comparison

Item	Baseline	Fixed
Git Commit	f3674e8	d5d1356
Timestamp	2025-11-20 09:42:40	2025-11-20 11:20:03
R Version	R version 4.3.2 (2023-10-31 ucrt)	R version 4.5.1 (2025-06-13 ucrt)
Issue Number	67	67

2.1 Global Parameters

Table 3: Global Parameter Comparison

	Parameter	Baseline	Fixed	Changed
WTP	WTP	51,000	51,000	No
time_horizon	time_horizon	520	520	No
n_samples	n_samples	5,000	5,000	No
n_sim	n_sim	5,000	5,000	No
cl	cl	0.01923077	0.01923077	No
dr	dr	0.04	0.04	No
DSA_mult	DSA_mult	0.2	0.2	No
USE_BOTH_MODELS	USE_BOTH_MODELS	0	0	No

3 Base Case Results Comparison

Table 4: Base Case Cost-Effectiveness Results Comparison

Strategy	Cost (€)				Effect (QALYs)				ICER (€/QALY)		
	Baseline	Fixed	Difference	% Change	Baseline	Fixed	Difference	% Change	Baseline	Fixed	Difference
control	21,748.34	21,748.34	0.00	0.00	1.8040	1.8040	0.0000	0.00	—	—	—
crp	55,992.46	55,992.46	0.00	0.00	1.9153	1.9153	0.0000	0.00	307,907	307,907	0
tlr	76,439.41	76,439.41	0.00	0.00	1.2957	1.2957	0.0000	0.00	—	—	—
tmb_braf	68,032.43	68,032.43	0.00	0.00	1.8235	1.8235	0.0000	0.00	—	—	—

3.1 Net Monetary Benefit Comparison

Table 5: Net Monetary Benefit Comparison at WTP = €51,000

Strategy		Baseline (€)	Fixed (€)	Difference (€)
1	control	70,257.69	70,257.69	0.00
2	crp	41,685.57	41,685.57	0.00
4	tmb_braf	24,963.72	24,963.72	0.00
3	tlr	-10,359.72	-10,359.72	0.00

Optimal Strategy: Unchanged (control)

4 PSA Results Comparison

4.1 Summary Statistics

Table 6: PSA Mean Costs and Effects (n = 5,000 simulations)

Strategy	Mean Cost (€)			Mean Effect (QALYs)		
	Baseline	Fixed	Difference	Baseline	Fixed	Difference
control	20,944.16	20,941.26	-2.90	1.7002	1.7029	0.0027
crp	54,047.81	54,129.95	82.14	1.6973	1.6998	0.0026
tlr	85,751.02	85,912.18	161.16	1.6217	1.6242	0.0025
tmb_braf	64,883.13	64,958.87	75.74	1.7946	1.7982	0.0036

4.2 95% Credible Intervals

Table 7: PSA 95% Credible Intervals

Strategy	Cost Baseline		Cost Fixed		Effect Baseline		Effect Fixed	
	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%
control	15,885.16	26,894.17	15,716.53	27,097.95	1.0970	2.3682	1.1074	2.3870
crp	40,703.95	69,247.36	40,616.81	69,830.12	1.1132	2.2747	1.1278	2.2465
tlr	61,436.17	114,517.64	61,239.96	114,479.34	1.0310	2.2031	1.0470	2.2050
tmb_braf	47,451.42	85,174.43	47,478.52	85,470.76	1.1060	2.5444	1.1243	2.5665

5 Visual Comparison

5.1 PSA Scatter Plots (Side-by-Side)

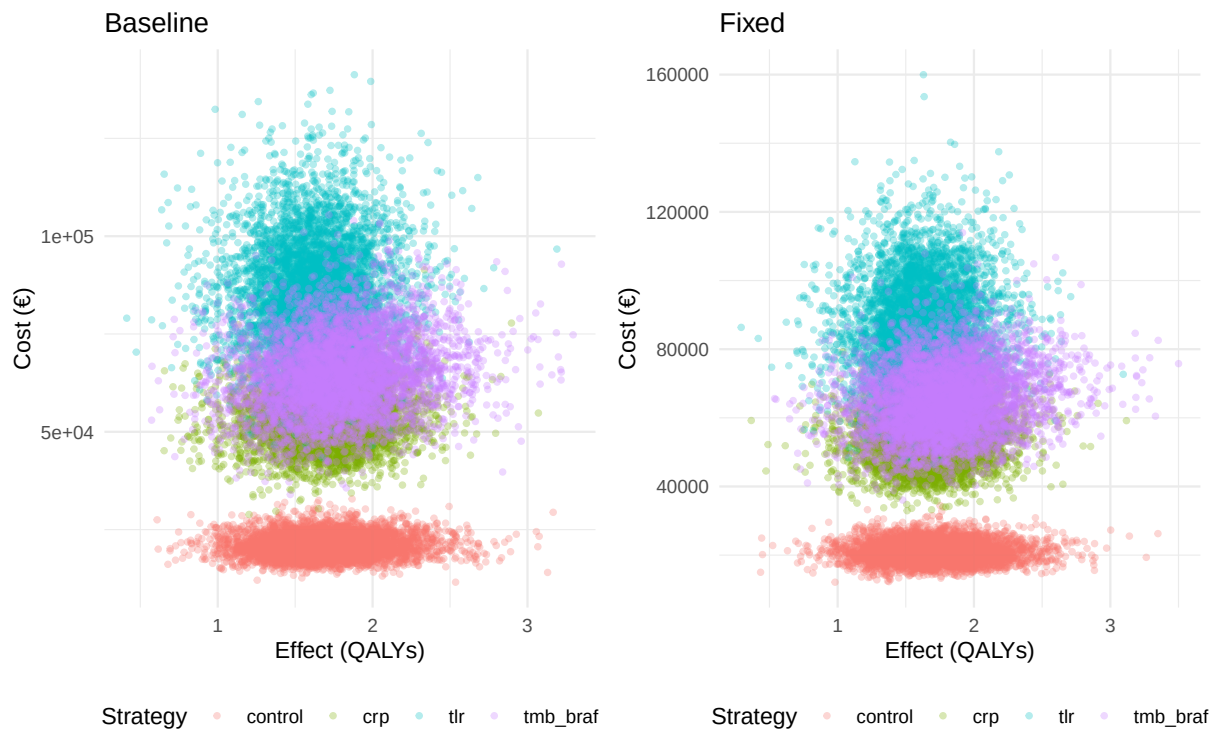


Figure 1: Cost-effectiveness plane showing PSA simulations baseline (left) and fixed (right)

5.2 PSA Scatter Plots (Overlaid)

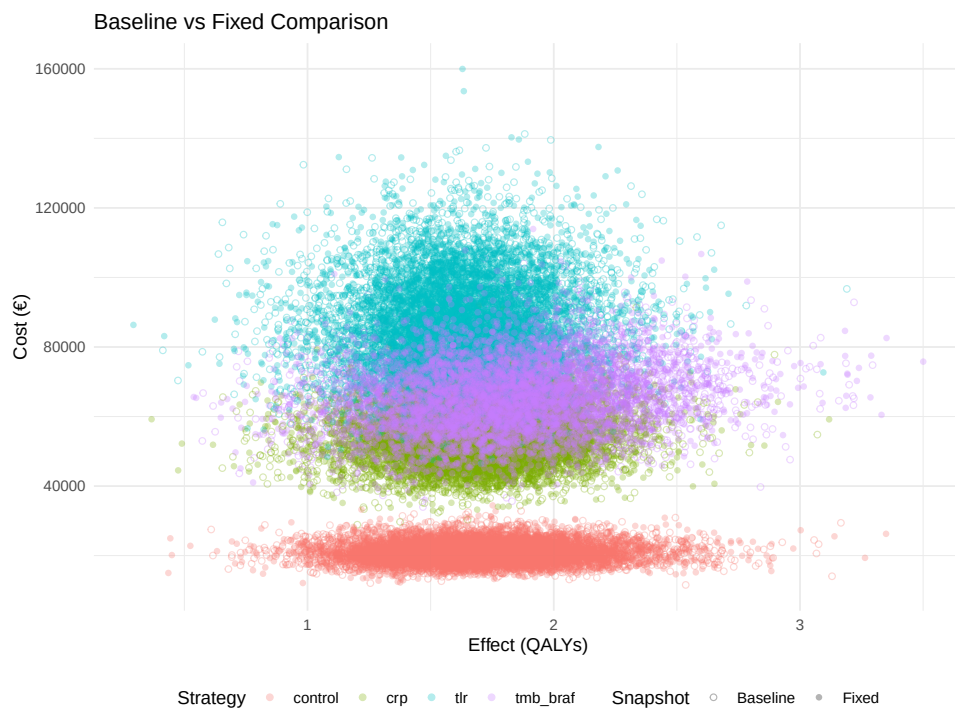


Figure 2: Cost-effectiveness plane with overlaid baseline and fixed PSA simulations

6 Comparison Across Recent Issues

This table summarizes the impact of bug fixes across the 3 most recent issues:

Table 8: Summary of Bug Fix Impacts (3 Most Recent Issues)

Issue	Title	Optimal (Baseline)	Optimal (Fixed)	Changed?	Max Cost Change (€)	Max Effect Change (QALYs)	Max ICER Change (€/QALY)
#67	—	control	control	No	0.00	0.0000	0
#62	—	control	control	No	0.00	0.0000	0
#64	—	control	control	No	0.00	0.0000	0

7 Summary

The comparison for issue #67 shows:

- **Optimal Strategy:** Unchanged (control)
- **Maximum Cost Change:** €0.00 (minor)
- **Maximum Effect Change:** 0.000 QALYs (minor)
- **Maximum ICER Change:** €0/QALY (minor)
- **Parameter Changes:** None detected

Interpretation: Changes are within expected bounds for bug fixes that improve numerical stability or code quality.

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